PENGWIN 2026 Task 1 — v3.4 TotalSegmentator-Initialized Candidate

Algorithm Description · 2026-07-28

1. Submission status

`v3.4-totalpretrain-t075` is an upload candidate, not an automatically activated replacement. The currently known `model_v3_0` package remains available as the rollback model until this candidate passes the Grand Challenge container test.

The candidate changes the two segmentation checkpoints and the Stage-B agglomeration threshold. It retains the validated v3.3 hybrid target-family router and the Grand Challenge input/output contract.

2. Method overview

The submission performs per-fragment instance segmentation of pelvic and femoral fractures using a serial two-stage STU-Net-B cascade:

1. **Stage A — anatomy.** A 5-class whole-volume model identifies sacrum, left/right hipbone and femur.
2. **Hybrid family routing.** A random-forest classifier selects pelvic or femur when confident. The organizers' official acquisition rule is used only when the RF is uncertain.
3. **Stage B — fracture.** A per-anatomy model emits four ABBC channels and nine learned same-instance affinity channels.
4. **Instance decoding.** Average-linkage agglomeration converts the affinity field into fragment instances and removes components below about 1 cm3.
5. **Assembly.** Instances are mapped to the official label ranges: sacrum 1-50, left hip 51-100, right hip 101-150 and femur 151-200.

3. Deployed model contract

Component	Deployed value
Stage A dataset	`Dataset539_PelvicFemurAnatomyV3`
Stage A trainer	`PengwinTrainerSTUNetBaseAnatomyV301`
Stage A fold/checkpoint	fold 0 / `checkpoint_best.pth`
Stage A initialization	from scratch on refreshed PENGWIN anatomy data
Router	v3.3 hybrid RF, confidence margin 0.15
Stage B dataset	`Dataset538_PelvicFemurBICMFragmentV5`
Stage B trainer	`PengwinTrainerSTUNetBaseAffinityV308DeployedVal`
Stage B fold/checkpoint	fold 0 / `checkpoint_best.pth`
Stage B initialization	official TotalSegmentator `base_ep4k`, then refreshed-data fine-tuning
Stage B output	13 channels: 4 ABBC + 9 affinity
Decoder	`decode_affinity_agglo`, `T=0.75`

Only Stage B is TotalSegmentator-initialized. Stage A is the refreshed-data scratch model. The Stage-B training process reached epoch 180 and then ended at the start of epoch 181 without producing `checkpoint_final.pth`; the valid `checkpoint_best.pth` selected during training is therefore the packaged artifact.

4. Model archive layout

Grand Challenge extracts the uploaded archive directly under `/opt/ml/model`. The tarball is created with `tar -C model_payload -czf model.tar.gz .`, producing:

```
/opt/ml/model/
├── nnunet/results/
│   ├── Dataset539_PelvicFemurAnatomyV3/
│   │   └── PengwinTrainerSTUNetBaseAnatomyV301__nnUNetResEncUNetLPlans__3d_fullres/
│   └── Dataset538_PelvicFemurBICMFragmentV5/
│       └── PengwinTrainerSTUNetBaseAffinityV308DeployedVal__nnUNetResEncUNetLPlans__3d_fullres/
└── stage1_router/stage1_target_router_fold0.joblib
```

The router artifact is serialized natively with scikit-learn 1.6.1, matching the container dependency. Its predictions were checked against the original scikit-learn 1.7.2 serialization on 4,096 deterministic inputs.

5. Local evaluation

All rows below use the same 68-case official-aligned evaluation set.

Experiment	Dice	HD95 (mm)	ASSD (mm)	Instance F1	Split fragments
Existing release	0.884121	3.857464	1.039815	**0.933582**	457
Refreshed-data scratch	0.865772	4.009136	0.927063	0.846223	537
v3.4 candidate, fixed `T=0.45`	**0.885355**	3.422187	0.800181	0.921553	456
v3.4 candidate, selected `T=0.75`	0.882088	**3.260951**	**0.774668**	0.930066	**426**

The selected threshold trades 0.00327 Dice relative to `T=0.45` for better HD95, ASSD, fragment over-splitting and instance F1. Relative to the existing release, it improves HD95 by 0.5965 mm, ASSD by 0.2651 mm and removes 31 split fragments, while Dice changes by -0.0020 and instance F1 by -0.0035. These are local proxy results, not hidden-test leaderboard results.

6. Preprocessing and inference

Images are canonicalized to LPS and processed through the nnU-Net plans used in training. Stage A predicts mutually exclusive anatomy masks. The selected anatomy ROI is converted to bone-window CT plus routing-derived context channels for Stage B. Inference runs the two models serially on one GPU, restores the prediction to the input geometry and writes one `/output/images/pelvic-fracture-segmentation/*.mha` label map.

7. Reproducibility and limitations

- Fold 0 is used for both stages; no ensemble is deployed.

- `T=0.75` was selected on the reported validation data, so an independent Grand Challenge container/hidden-test check is required before activation.
- Stage B has no `checkpoint_final.pth` because the run ended after epoch 180; the packaged checkpoint is the valid `checkpoint_best.pth`.
- The official challenge evaluator and hidden test distribution are not local, so the reported metrics are comparative proxies.
- Exact artifact hashes and source paths are recorded in the release `MODEL_MANIFEST.json`.

8. Runtime and licensing

The Docker image targets an NVIDIA T4 16 GiB GPU. Stage A and Stage B are loaded serially to limit peak memory. STU-Net-B and nnU-Net are used under their respective open-source licenses; the Stage-B initialization comes from the official TotalSegmentator `base_ep4k` checkpoint.

9. References

- Isensee, F. et al. nnU-Net, *Nature Methods* 18, 203-211 (2021).
- STU-Net / TotalSegmentator bone pretraining.
- PENGWIN 2026 Task 1 baseline: `github.com/YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline`.